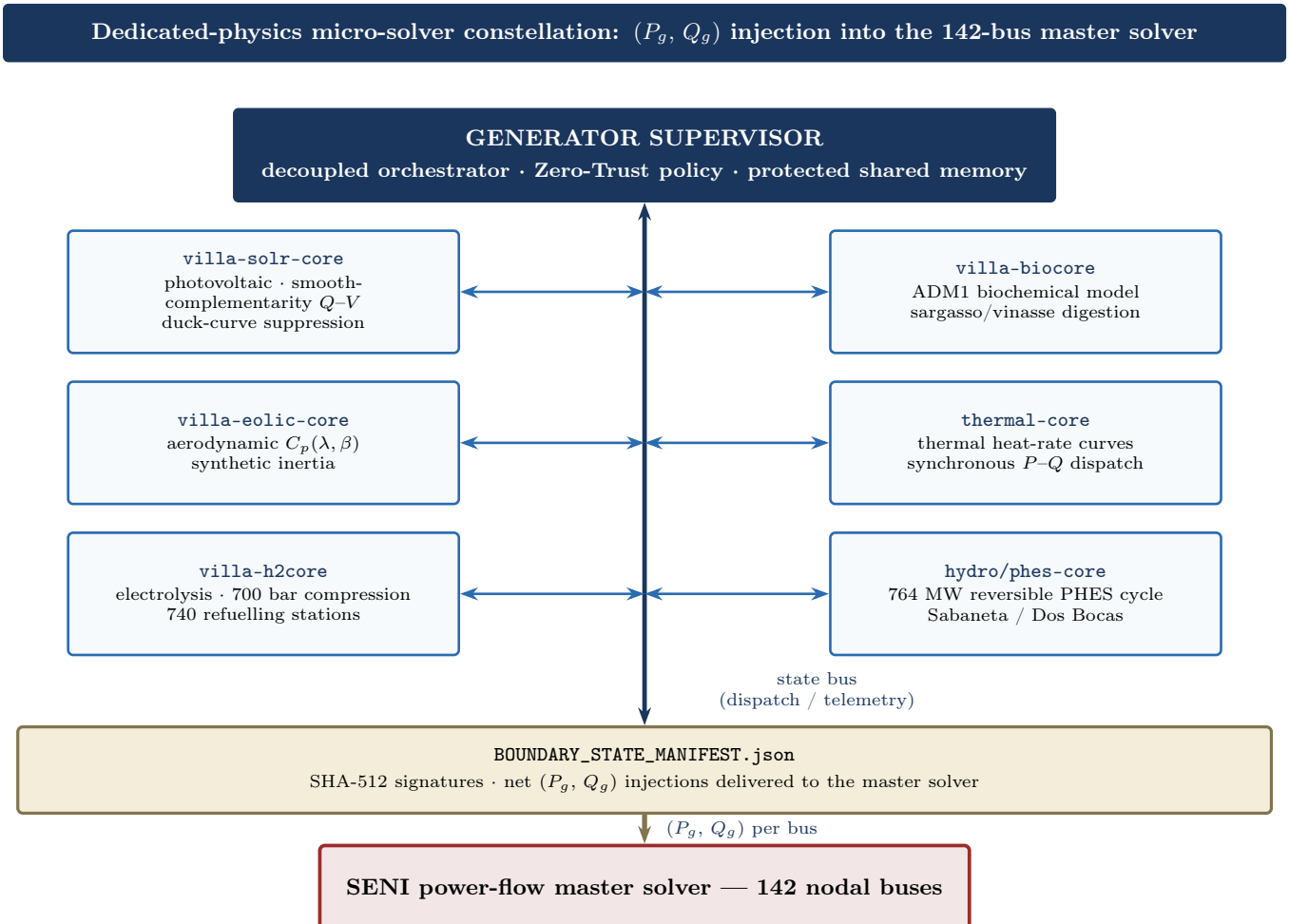

Dedicated-physics micro-solver constellation: (P_g, Q_g) injection into the 142-bus master solver
GENERATOR SUPERVISOR
decoupled orchestrator · Zero-Trust policy · protected shared memory
villa-solr-core
photovoltaic · smooth-complementarity Q–V
duck-curve suppression
villa-biocore
ADM1 biochemical model
sargasso/vinasse digestion
villa-eolic-core
aerodynamic $C_p(\lambda, \beta)$
synthetic inertia
thermal-core
thermal heat-rate curves
synchronous P–Q dispatch
villa-h2core
electrolysis · 700 bar compression
740 refuelling stations
hydro/phes-core
764 MW reversible PHES cycle
Sabaneta / Dos Bocas
state bus
(dispatch / telemetry)
BOUNDARY_STATE_MANIFEST.json
SHA-512 signatures · net (P_g, Q_g) injections delivered to the master solver
(P_g, Q_g) per bus
SENI power-flow master solver — 142 nodal buses